Module 5: Internetworking, TCP/IP & Routing

IPv4 VLSM Subnetting, TCP Congestion Control, Dijkstra & DVR

Module V: Internetworking, TCP/IP & Routing — Topics Covered

1. IPv4 Addressing, Classless CIDR & VLSM Subnetting
2. TCP 3-Way Handshake & Congestion Control
3. Dijkstra Shortest Path (OSPF) & Bellman-Ford (RIP)
4. Application Protocols: DNS, DHCP, HTTP/2, SMTP

1. TCP Congestion Control Mechanics

TCP maintains a congestion window ($cwnd$) and slow-start threshold ($ssthresh$):

- **Slow Start:** $cwnd$ starts at $1\ \mathrm{MSS}$ and doubles every RTT ($cwnd = cwnd \times 2$) exponentially until $cwnd \geq ssthresh$.
- **Congestion Avoidance:** $cwnd$ increases linearly by $1\ \mathrm{MSS}$ per RTT (Additive Increase).
- **Triple Duplicate ACKs (Fast Retransmit & Recovery):** $ssthresh = cwnd/2, cwnd = ssthresh + 3\ \mathrm{MSS}$, retransmit lost segment immediately.
- **Timeout:** $ssthresh = cwnd/2, cwnd = 1\ \mathrm{MSS}$, re-enter Slow Start.

🏛 **BIT Mesra VLSM Subnetting Problem (10 Marks)**

Problem: Given network block `192.168.1.0/24`, design subnets for Dept A (100 hosts), Dept B (50 hosts), and Dept C (25 hosts).

Solution:

- **Dept A (100 hosts):** Needs $2^7 - 2 = 126$ IPs $\implies /25$. Subnet: `192.168.1.0/25` (Range: `.1` to `.126`, Broadcast: `.127`).
- **Dept B (50 hosts):** Needs $2^6 - 2 = 62$ IPs $\implies /26$. Subnet: `192.168.1.128/26` (Range: `.129` to `.190`, Broadcast: `.191`).
- **Dept C (25 hosts):** Needs $2^5 - 2 = 30$ IPs $\implies /27$. Subnet: `192.168.1.192/27` (Range: `.193` to `.222`, Broadcast: `.223`).